

The Superstore sales dataset will be analyzed to determine if sales can be predicted based on the timing of customers orders. The dataset includes variables such as order dates, ship dates, customers, product information and sales figure. After reviewing the dataset structure, I will attempt to identify missing and problematic values. Clean the order date variable and create the days variable representing the number of days of sales over time. Using Python, a scatter plot and a histogram will be used to examine the linear relationship between time and sales. An evaluation of the model will be done using performance metrics such as R-squared, mean absolute error (MAE) and root mean square error (RMSE). Finally the results will be interpreted and discussed to determine if time is a useful predictor of Superstore sales.

```
from google.colab import files  
  
uploaded = files.upload()
```

[Show hidden output](#)

```
import pandas as pd

df = pd.read_csv('train.csv')

df.head()
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment
0	1	CA-2017-152156	08/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer
1	2	CA-2017-152156	08/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer
2	3	CA-2017-138688	12/06/2017	16/06/2017	Second Class	DV-13045	Darrin Van Huff	Corporate
3	4	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O'Donnell	Consumer
4	5	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O'Donnell	Consumer

```
df.columns
```

```
Index(['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode',  
      'Customer ID', 'Customer Name', 'Segment', 'Country', 'City',  
      'State',  
      'Postal Code', 'Region', 'Product ID', 'Category', 'Sub-  
Category',  
      'Product Name', 'Sales'],  
      dtype='object')
```

```
df.describe()
```

```
df['Order Date'] = pd.to_datetime(df['Order Date'], dayfirst=True)  
  
df['Days'] = (df['Order Date'] - df['Order Date'].min()).dt.days  
  
df[['Order Date', 'Days', 'Sales']].head()
```

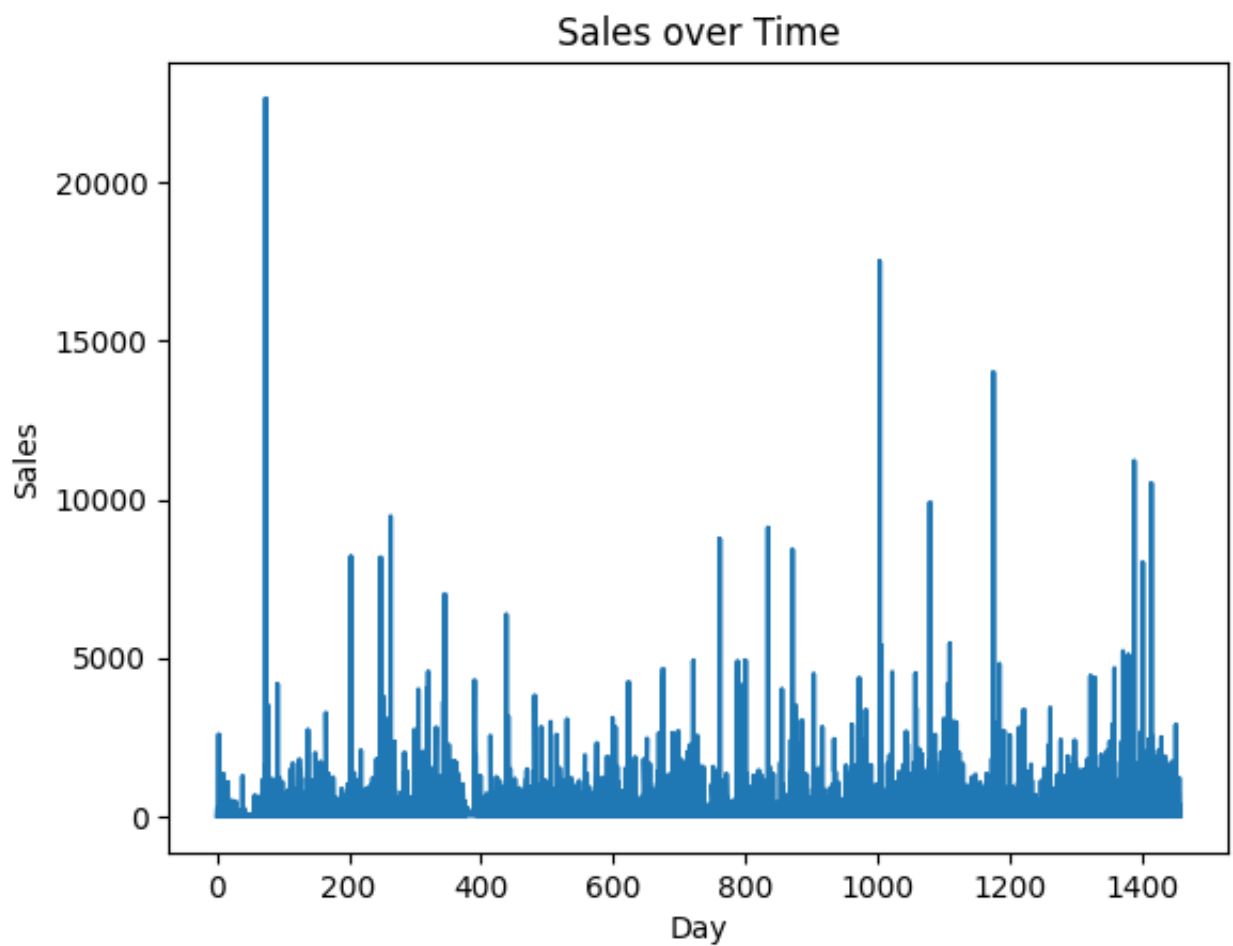
	Order Date	Days	Sales
0	2017-11-08	1040	261.9600
1	2017-11-08	1040	731.9400
2	2017-06-12	891	14.6200
3	2016-10-11	647	957.5775
4	2016-10-11	647	22.3680

```
import matplotlib.pyplot as plt

df = df.sort_values('Days')

x=df['Days']
y=df['Sales']

plt.plot(x,y)
plt.xlabel('Day')
plt.ylabel('Sales')
plt.title('Sales over Time')
plt.show()
```



```
import numpy as np
import matplotlib.pyplot as plt

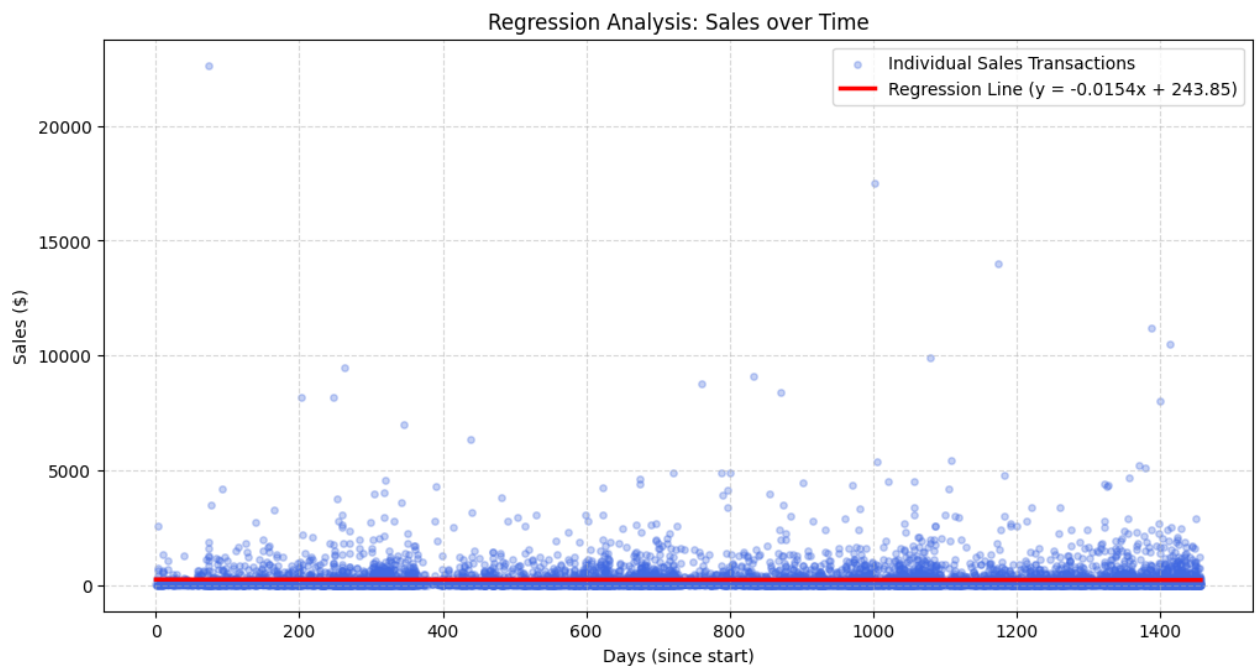
x = df['Days']
y = df['Sales']
```

```
m, c = np.polyfit(x, y, 1)

plt.figure(figsize=(12, 6))
plt.scatter(x, y, alpha=0.3, color='royalblue', label='Individual Sales')

plt.plot(x, m*x + c, color='red', linewidth=2.5, label=f'Regression Line')

plt.xlabel('Days (since start)')
plt.ylabel('Sales ($)')
plt.title('Regression Analysis: Sales over Time')
plt.legend()
plt.grid(True, linestyle='--', alpha=0.5)
plt.show()
```



```
from sklearn.linear_model import LinearRegression
```

```

from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt

lr_model = LinearRegression()
lr_model.fit(X_train, y_train)

lr_predictions = lr_model.predict(X_test)

lr_mae = mean_absolute_error(y_test, lr_predictions)
lr_r2 = r2_score(y_test, lr_predictions)

print("=== Linear Regression Model Evaluation ===")
print(f"Mean Absolute Error (MAE): ${lr_mae:.2f}")
print(f"R-squared Score: {lr_r2:.4f}")

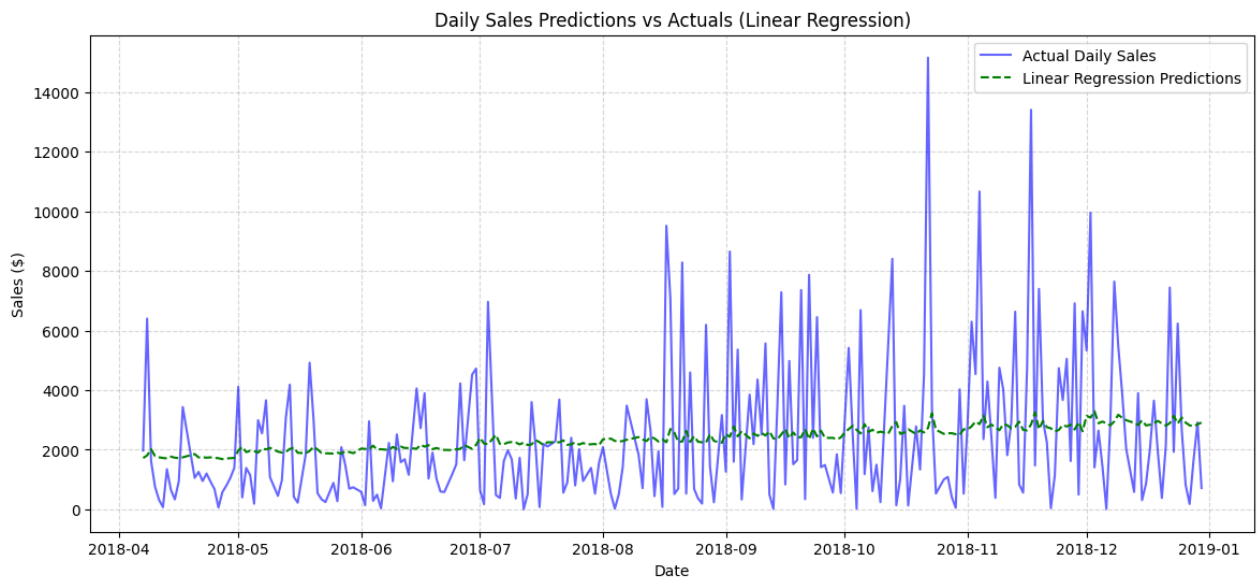
plt.figure(figsize=(14, 6))
plt.plot(daily_df['Order Date'].iloc[split_idx:], y_test, label='Actual')
plt.plot(daily_df['Order Date'].iloc[split_idx:], lr_predictions, label='Predictions')
plt.title('Daily Sales Predictions vs Actuals (Linear Regression)')
plt.xlabel('Date')
plt.ylabel('Sales ($)')
plt.legend()
plt.grid(True, linestyle='--', alpha=0.5)
plt.show()

```

```

=== Linear Regression Model Evaluation ===
Mean Absolute Error (MAE): $1763.73
R-squared Score: 0.0599

```



```
from sklearn.metrics import mean_squared_error
import numpy as np

rmse = np.sqrt(mean_squared_error(y_test, lr_predictions))
print(f'Root Mean Squared Error (RMSE): ${rmse:.2f}')
```

Root Mean Squared Error (RMSE): \$2378.19

Double-click (or enter) to edit

The final analysis used a linear regression model to examine whether the timing of orders could predict sales in the Superstore dataset. Data cleaning and preparation were completed before the analysis. A scatter plot and histogram were used to explore the data and identify unusually high sales values. The model produced a Mean Absolute Error (MAE) of 1,763.73, meaning that predictions differed from actual sales by approximately 1,764 on average. The R-squared value was 0.0599 showing the model explained 6% of the variation in sales. The Root Mean Squared Error. Showing large predictive errors. This model shows that timing alone is not an accurate predictor of sales. Other variables and regression models may be more suited.

Gemini was used in the coding of this assignment.

